

NORM PRINCIPLE FOR SEMISIMPLE GROUPS OVER FUNCTION FIELDS OF CURVES

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ABSTRACT. We prove that the norm principle holds for any semisimple linear algebraic group G of type D_n over function fields of smooth, projective, geometrically integral curves over p -adic fields, and also over function fields of smooth, projective, geometrically integral curves over number fields, in the case where G is induced from the underlying number field. Our method is based on local–global principles and the Rost invariant.

Let L/K be a finite separable field extension, G be a semisimple linear algebraic group defined over K , and Z be the center of G . Consider the following diagram

$$\begin{array}{ccc} H^1(L, Z) & \xrightarrow{\alpha_L} & H^1(L, G) \\ \text{\scriptsize $cor_{L/K}$} \downarrow & & \\ H^1(K, Z) & \xrightarrow{\alpha_K} & H^1(K, G), \end{array}$$

where $cor_{L/K}$ denotes the corestriction map in Galois cohomology, and the map α is induced by the inclusion $Z \hookrightarrow G$.

We say that the norm principle holds for G over K , if for every finite separable field extension L/K , we have $cor_{L/K}(\ker \alpha_L) \subseteq \ker \alpha_K$. We remark that there are two variants of the norm principle: the H^0 -variant and the H^1 -variant (following the notation in [24] and [25]). In this article, we work with the H^1 -variant of the norm principle, and we simply call it the *norm principle*. There is an equivalence between the two variants [25, Theorem 4.3.1].

It is an open question whether the norm principle holds in general for every semisimple linear algebraic group G over any field K .

Suppose that $\text{char}(K) \neq 2$, $cd(K) \leq 2$, and G is simply connected of classical type. Bayer–Parimala’s proof of Serre’s conjecture II shows that the Galois cohomology set $H^1(K, G)$ is trivial ([2]). Therefore, the norm principle automatically holds for G over K . Also, it is known that the norm principle holds for all semisimple groups with no components of type $D_n, n \geq 4$, E_6 , or E_7 , over arbitrary fields (see [1, Theorem 1.1]).

In this paper, we study the norm principle for groups of type D_n over fields K with $cd_2(K) \leq 3$ and $u(K) \leq 10$ using the Rost invariant. This class includes, in particular, function fields of curves over p -adic fields. Using a local–global approach, we prove the following main result:

Theorem: (2.3) Let k be a number field, and C be a smooth, projective, geometrically integral curve over k with function field $K = k(C)$. Let G be a semisimple group of type D_n (with D_4 non-trivialitarian) defined over k . Then, the norm principle holds for G over K .

1. ROST INVARIANT AND THE NORM PRINCIPLE

The norm principle for a simply connected cover of a semisimple group G implies the norm principle for G (see [24, Lemma 2.3]). Therefore, it is sufficient to prove the norm principle for all semisimple, simply connected groups.

Let K be a field and G be a semisimple, simply connected linear algebraic group defined over K . There is an invariant due to Rost,

$$R : H^1(K, G) \rightarrow H^3(K, \mathbb{Q}/\mathbb{Z}(2))$$

which is functorial in K (see [15, Section 31] for details). The first step of our strategy in this paper to study the norm principle, is to establish a relation between kernel of the Rost invariant for G and the norm principle for G .

Theorem 1.1. *Let K be a field, and G be a semisimple, simply connected linear algebraic group defined over K , with Rost invariant R . Assume that the kernel of R_K is trivial. Then the norm principle holds for G over K .*

Proof. Let Z be the center of G , and $\alpha : H^1(-, Z) \rightarrow H^1(-, G)$ the Galois cohomology map induced by the map $Z \rightarrow G$. Consider the following maps:

$$\begin{array}{ccccc} H^1(K, Z) & \xrightarrow{\alpha_K} & H^1(K, G) & \xrightarrow{R_K} & H^3(K, \mathbb{Q}/\mathbb{Z}(2)). \\ & & \searrow & \nearrow & \\ & & \iota_K & & \end{array}$$

Let L be a finite separable field extension of K . Consider the following diagram, where the vertical maps are the corestriction maps in Galois cohomology:

$$(1.1) \quad \begin{array}{ccccc} & & \iota_L & & \\ & \searrow & & \nearrow & \\ H^1(L, Z) & \xrightarrow{\alpha_L} & H^1(L, G) & \xrightarrow{R_L} & H^3(L, \mathbb{Q}/\mathbb{Z}(2)) \\ \downarrow \text{cor}_{L/K} & & & & \downarrow \text{cor}_{L/K} \\ H^1(K, Z) & \xrightarrow{\alpha_K} & H^1(K, G) & \xrightarrow{R_K} & H^3(K, \mathbb{Q}/\mathbb{Z}(2)). \\ & \nearrow & & \searrow & \\ & & \iota_K & & \end{array}$$

Let Z° be the dual multiplicative group scheme of Z . There is a natural cup product $H^1(K, Z) \otimes H^2(K, Z^\circ) \rightarrow H^3(K, \mathbb{Q}/\mathbb{Z}(2))$ (see [10, Theorem 1.2]). Let t_G be the Tits class of G , which is an element in $H^2(K, Z)$ (see [15, Page 426] for the definition of the Tits class). There is a natural map $\hat{\rho}^* : H^2(K, Z) \rightarrow H^2(K, Z^\circ)$ (see [10, Section 5.C] for the details of the map $\hat{\rho}^*$). Let $s_G = \hat{\rho}^*(t_G)^{-1}$. In [10, Theorem 1.2], it is proved that the map ι_K is given by the formula

$$\iota_K(x) = x \cup s_G, \text{ for every } x \in H^1(K, Z).$$

The Rost invariant is preserved under scalar extension. Therefore, we have the formula

$$\iota_L(x) = x \cup s_G, \text{ for every } x \in H^1(L, Z),$$

where $\text{res}_{L/K}$ denotes the restriction map.

Let $u \in H^1(L, Z)$. We have

$$\begin{aligned} \text{cor}_{L/K}(\iota_L(u)) &= \text{cor}_{L/K}(u \cup \text{res}_{L/K}(s_G)) \\ &= \text{cor}_{L/K}(u) \cup s_G \quad (\text{by projection formula in Galois cohomology}) \\ &= \iota_K(\text{cor}_{L/K}(u)). \end{aligned}$$

Note that s_G is actually defined over K , which allows us to use the projection formula.

The above computation shows that the Diagram (1.1) is commutative. Let $u \in \ker \alpha_L$. This implies that $u \in \ker \iota_L$, hence $\text{cor}_{L/K}(\iota_L(u))$ is trivial in $H^3(K, \mathbb{Q}/\mathbb{Z}(2))$. By commutativity of Diagram (1.1), $\text{cor}_{L/K}(u) \in \ker \iota_K$. By assumption, $\ker R_K$ is trivial. Therefore, $\text{cor}_{L/K}(u) \in \ker \alpha_K$, and the norm principle holds for G over L/K . $\square$

The norm principle for a semisimple group can be obtained from the norm principle for all simple components of it (see [1, Proposition 5.2]). The norm principle is open in general for semisimple groups of type D_n , E_6 , and E_7 .

We recall that the u -invariant of a field K , denoted by $u(K)$, is the largest dimension of an anisotropic quadratic form over K and is ∞ if such an integer does not exist for K .

The following theorem proves the norm principle for several types of groups, under assumptions on the base field. We remark that part (c) of the theorem is known; however, we give a new proof.

Theorem 1.2. *The norm principle holds for the group G over K , in each of the following cases:*

- (a) *Let K be a field of characteristic $\neq 2$ with $\text{cd}_2(K) \leq 3$. Suppose further that $u(K) \leq 10$. Let $G = \text{Spin}(A, \sigma)$, where (A, σ) is a central simple algebra with $\text{ind}(A) \leq 4$, $\text{deg}(A) = 2n$ for $n \geq 2$, and orthogonal involution defined over K . In particular, K can be the function field of any smooth projective geometrically integral curve over a p -adic field.*
- (b) *Let K be a perfect field of char $\neq 2, 3$, and G be a simple, simply connected quasi-split group of type E_6 , E_7 , or ${}^{3,6}D_4$ defined over K .*
- (c) *Let n be a natural number such that $n + 1$ is square-free, and K be a field of characteristic coprime to $n + 1$. Let A be a central simple algebra with degree $n + 1$ defined over K , and $G = \text{SL}_1(A)$.*

Proof. (a) If $G = \text{Spin}(A, \sigma)$ is defined over K with $\text{char } K \neq 2$, $\text{cd}_2(K) \leq 3$, $u(K) \leq 10$, and $\text{ind}(A) \leq 4$, then by [21, Theorem 1.8], the Rost invariant of $G = \text{Spin}(A, \sigma)$ has trivial kernel over K . The norm principle for G follows from Theorem 1.1.

If K is the function field of a smooth projective geometrically integral curve defined over a p -adic field, then K has cohomological dimension 3. Furthermore, $u(K) = 8$ (see [20]). Hence the norm principle holds for G over K .

(b) The Rost invariant of G has trivial kernel by [6, Theorem 6.1 and Theorem 6.12]. The result then follows from Theorem 1.1.

(c) We have $H^1(K, G) = K^*/\text{Nrd}(A^*)$. The formula for the Rost invariant of G is then given by the following: for every element $x \in K^*$,

$$R_K(x\text{Nrd}(A^*)) = [x] \cup [A].$$

Theorem 12.2 in [17] shows that $[x] \cup [A]$ is trivial if and only if x is a reduced norm. Therefore, the Rost invariant of G has trivial kernel. The result follows from Theorem 1.1. $\square$

Corollary 1.3. *Let K be the function field of any smooth projective geometrically integral curve over a p -adic field. A theorem of Saltman [22] and a result of Parimala and Suresh [19, Theorem 1], show that if A is a central simple algebra over K with exponent of $A \leq 2$, then $\text{ind}(A) \leq 4$. The algebra A in part (a) of Theorem 1.2 has exponent at most 2, because it admits an orthogonal involution. Therefore, by part (a) of Theorem 1.2, the norm principle holds for any simple, simply connected (and hence any semisimple) linear algebraic group of classical type D_n , where $n \geq 2$, over K .*

Remark 1.4. *The norm principle for groups of outer type A_n , and groups of type B_n and C_n is proved over arbitrary fields in [1]. The kernel of the Rost invariant is not necessarily trivial for these groups over arbitrary fields K . However, with certain assumptions on the field K - in particular on the 2-cohomological dimension of K - and on the group G , the Rost kernel of the simply connected cover of G is known to be trivial ([21, Theorems 1.7 and 1.8]), hence the norm principle holds for G over K , under each of the following assumptions:*

- *When K is a field of characteristic $p \geq 0$ with $p \neq 2$, $\text{cd}_2(K) \leq 3$, and G is of outer type A_n , i.e., $G = \text{SU}(B, \tau)$, where B is a central simple algebra of degree $n + 1$, for $n \geq 2$, over a quadratic extension F over K with a unitary $F|K$ involution τ . Further, the index of B is square-free and coprime to p , if $p \geq 0$.*
- *When K is a field of characteristic $\neq 2$ with $\text{cd}_2(K) \leq 3$ and $u(K) \leq 10$, and G is any group of type B_n over K .*
- *When K is a field of characteristic $p \geq 0$ with $p \neq 2$, $\text{cd}_2(K) \leq 3$, and G is of type C_n , i.e., $G = \text{Sp}(A, \sigma)$ for a central simple algebra A over K of degree $2n$, for $n \geq 1$, with a symplectic involution. Further, $\text{ind}(A) \leq 4$.*

For all the cases above, the norm principle for G over K follows directly from Theorem 1.1.

2. THE NORM PRINCIPLE FOR GROUPS OF TYPE D_n OVER FUNCTION FIELDS OF CURVES OVER NUMBER FIELDS

Let K be a field with a collection of overfields $\{K_\nu\}_{\nu \in \Omega}$. Let G be a semisimple linear algebraic group defined over K . The next theorem asserts that if the norm principle holds for G over a collection of overfields $\{K_\nu\}_{\nu \in \Omega}$ of K , and a local-global principle is valid, then the norm principle holds for G over K . We use this fact to prove the norm principle for certain groups of type D_n over function fields of curves over number fields (see Theorem 2.3). Note that the validity of the norm principle for semisimple groups of type D_n over function fields of curves over number fields is an open problem in general.

Lemma 2.1. *Let K be a field, with a collection of overfields $\{K_\nu\}_{\nu \in \Omega}$. Let G be a semisimple linear algebraic group defined over K . Assume that the kernel of the global to local map*

$$\theta_K : H^1(K, G) \longrightarrow \prod_{\nu \in \Omega} H^1(K_\nu, G)$$

is trivial. Furthermore, assume that the norm principle holds for G over K_ν , for each $\nu \in \Omega$. Then the norm principle holds for G over K .

Proof. Let Z be the center of G , and let $\alpha : H^1(-, Z) \rightarrow H^1(-, G)$ be the Galois cohomology map induced by the embedding $Z \rightarrow G$. Let L/K be a finite separable field extension. Consider the following commutative diagram:

$$\begin{array}{ccccc}
H^1(L, Z) & \xrightarrow{\alpha_L} & H^1(L, G) & & \\
\downarrow \text{cor}_{L/K} & \searrow \gamma_L & & \searrow \theta_L & \\
& \prod_{\nu \in \Omega} H^1(K_\nu \otimes_K L, Z) & \xrightarrow{\beta_L} & \prod_{\nu \in \Omega} H^1(K_\nu \otimes_K L, G) & \\
& \downarrow \text{cor}_{K_\nu \otimes_K L/K_\nu} & & & \\
H^1(K, Z) & \xrightarrow{\alpha_K} & H^1(K, G) & & \\
& \searrow \gamma_K & & \searrow \theta_K & \\
& \prod_{\nu \in \Omega} H^1(K_\nu, Z) & \xrightarrow{\beta_K} & \prod_{\nu \in \Omega} H^1(K_\nu, G) &
\end{array}$$

Let $u \in \text{Ker } \alpha_L$, and $(w_\nu)_{\nu \in \Omega} \in \prod_{\nu \in \Omega} H^1(K_\nu \otimes_K L, Z)$ be the image of u under γ_L . Hence $(w_\nu)_{\nu \in \Omega} \in \text{Ker } \beta_L$. By assumption, the norm principle holds for G over each extension $K_\nu \otimes_K L/K_\nu$, therefore

$$\text{cor}_{K_\nu \otimes_K L/K_\nu}(w_\nu)_{\nu \in \Omega} \in \text{Ker } \beta_K.$$

Since

$$\theta_K \circ \alpha_K \circ \text{cor}_{L/K}(u) = \beta_K(\text{cor}_{K_\nu \otimes_K L/K_\nu}(w_\nu)_{\nu \in \Omega})$$

and $\text{Ker } \theta_K$ is by assumption trivial, we have $\text{cor}_{L/K}(u) \in \text{Ker } \alpha_K$. $\square$

Note that in [25, Theorem 1.3.2], a result similar to Lemma 2.1 is proved; however, in that setting, the field K is taken to be the function field of a curve over a number field, and the overfields K_ν are specifically the completions of K with respect to the valuations associated to the closed points on the curve. In Lemma 2.1, we do not assume K and K_ν are necessarily of this form.

Let k be a number field, and C be a smooth, projective, geometrically integral curve over k , with function field $K = k(C)$. We introduce two strategies for studying the norm principle for semisimple groups G over K . Both of these approaches utilize Lemma 2.1. But the overfields K_ν are different: In the first approach, the overfields K_ν are the fields obtained by completing the underlying number field k with respect to the places of k . In the second approach, the overfields K_ν are obtained by completing the field K with respect to divisorial valuations on K itself.

The first approach is as follows. Let k be a number field, and C be a smooth, projective, geometrically integral curve over k . Let $K = k(C)$ be the function field of C . We denote by Ω_k the set of places of k , and for $\nu \in \Omega_k$, let $K_\nu = k_\nu(C)$ denote the function field of the curve C_{k_ν} , which is an example of a semiglobal field.

In [8], Colliot-Thélène conjectures that for any semisimple, simply connected linear algebraic group G defined over K , the following map of pointed sets has trivial kernel:

$$H^1(K, G) \rightarrow \prod_{\nu \in \Omega_K} H^1(K_\nu, G).$$

This conjecture has been studied in various cases. For instance, Parimala and Preeti have proved the conjecture in some cases where the group G is in fact defined over k (see [18, Theorem 6.5]).

Remark 2.2. *Assume that the norm principle holds for G over every K_ν , for every $\nu \in \Omega_K$, and also assume that Colliot-Thélène’s conjecture holds for G over K . Then the norm principle holds for G over K . This is an immediate consequence of Lemma 2.1.*

Theorem 2.3. *Let k and K be as above. Let G be a semisimple classical linear algebraic group of type D_n , $n \geq 2$, defined over k . Then, the norm principle holds for G over K .*

Proof. It suffices to prove the norm principle for simple, simply connected groups G of type D_n over K , where G is defined over k . So without loss of generality we assume $G = \text{Spin}(A, \sigma)$, where A is a central simple algebra of degree $2n$ over K with orthogonal involution σ (these are the simply connected covers of groups of type D_n). Any such group $G = \text{Spin}(A, \sigma)$ can be realized as $G = \text{Spin}(h)$, where h is an n -dimensional hermitian form defined over (D, σ') , where (D, σ') is a central simple algebra over K , Brauer equivalent to A , with orthogonal involution σ' . By [18, Theorem 6.5], Colliot-Thélène’s conjecture holds for $\text{Spin}(h)$. Also the norm principle holds for $\text{Spin}(h)$ over each overfield K_ν by Corollary 1.3. Then the result follows from Remark 2.2. $\square$

Now we introduce the second framework for studying the norm principle for semisimple groups defined over function fields of algebraic curves over number fields, which differs from the one suggested by Remark 2.2. However, it still relies on a local-global approach and makes essential use of Lemma 2.1.

We propose the following question, which can be regarded as a Hensel’s lemma-type statement for the norm principle.

Question 2.4. *Let M be a complete discretely valued field with residue field $\overline{M}$. Assume that the norm principle holds for all simply connected linear algebraic groups defined over all finite separable extensions of $\overline{M}$. Does the norm principle hold for all simply connected linear algebraic groups defined over M ?*

Recall that, since the norm principle for a semisimple group follows from that of its simply connected cover, one may replace “simply connected” with “semisimple” in the formulation of Question 2.4 without loss of generality.

An affirmative answer to Question 2.4 implies that the norm principle holds for semisimple groups over a large class of complete discretely valued fields, including all complete discretely valued fields whose residue fields are number fields, p -adic fields, or global fields of positive characteristic: over all these residue fields the norm principle is known for all semisimple groups; see [12] for the case of number fields. For any p -adic field or any global field of positive characteristic k , the first Galois cohomology set $H^1(k, G)$ is trivial for any simply connected semisimple group G defined over k (see [14] and [13]), hence the norm principle holds for any semisimple group over k .

One natural example of such complete discretely valued fields is the completion of the function field of an algebraic curve with respect to valuations associated to closed points of the curve, when the curve is defined over a number field or a p -adic field.

Now we propose another question, similar to the question above, which concerns only groups of type D_n . Recall that any simple, simply connected linear algebraic group of type D_n over a field of characteristic $\neq 2$, is the spinor group $Spin(D, \sigma)$ for a central simple algebra with orthogonal involution. We also recall that any central simple algebra which admits an orthogonal involution has to be of index 2^r , for some nonnegative integer r .

Question 2.5. *Let M be a complete discretely valued field with residue field $\overline{M}$ with $\text{char}(\overline{M}) \neq 2$, and let r be a nonnegative integer. Assume that for all central simple algebras $\mathfrak{D}$ of index $\leq 2^r$ with orthogonal involution σ , defined over all finite separable field extensions F of $\overline{M}$, the norm principle holds for $Spin(\mathfrak{D}, \sigma)$ over F . Does the norm principle hold for $Spin(D, \sigma)$ over M , for all central simple algebras with orthogonal involution (D, σ) of index $\leq 2^r$ defined over M ?*

For $r = 0$, i.e. when D (resp. $\mathfrak{D}$) splits, the main result proved by Bhaskhar, Chernousov, and Merkurjev in [5, Theorem 5.1] gives an affirmative answer to Question 2.5. Note that the main result in [5] is stated in terms of the H^0 -variant of the norm principle. See [5, Lemma 2.2] for the equivalent statement in terms of the H^1 -variant of the norm principle for $Spin(q)$.

For $r = 1$, i.e. when D (resp. $\mathfrak{D}$) is a quaternion algebra, then Question 2.5 has affirmative answer by [24, Theorem 7.1].

To the best of our knowledge, there are no results in the literature concerning Question 2.5 for $r \geq 2$. Moreover, no analogous results appear to be available for groups of type E_6 and E_7 .

Now we discuss the second strategy for studying the norm principle for semisimple groups over function fields of smooth projective geometrically integral curves over number fields. Recall that it is sufficient to settle the norm principle for simple, simply connected groups.

We recall the following open question (see [5, Section 7]. For the details about divisorial valuations over function field of a curve, see for instance [7]): Let k be a number field, K be the function field of a smooth projective, geometrically integral curve over k , and G be a simply connected linear algebraic group over K . Let Ω be a set of divisorial valuations of K and for every $\nu \in \Omega$, let K_ν be the completion of K with respect to ν . Then, is the kernel of the natural map

$$\rho_{G, \Omega} : H^1(K, G) \longrightarrow \prod_{\nu \in \Omega} H^1(K_\nu, G)$$

trivial?

Theorem 2.6. *Let k , K , G , and Ω be as above. Assume that*

- (1) *For every $\nu \in \Omega$, Question 2.4 has affirmative answer for K_ν , and*
- (2) *kernel of the map $\rho_{G, \Omega}$ is trivial.*

Then, the norm principle holds for G over K .

Proof. The residue field $\overline{K}_\nu$ of each K_ν is either a finite extension of k (if the valuation ν is trivial on k), or a global field of positive characteristic (if ν is not trivial on k).

If $\overline{K}_\nu$ is a number field, then the norm principle holds for every semisimple group over $\overline{K}_\nu$ by [12]. If $\overline{K}_\nu$ is a global field of positive characteristic, then the Galois cohomology set $H^1(\overline{K}_\nu, \mathcal{G})$ is trivial for any semisimple, simply connected group $\mathcal{G}$ defined over $\overline{K}_\nu$ (see [14] and [13]), hence the norm principle holds for any semisimple group $\mathcal{G}$ over $\overline{K}_\nu$. Therefore, in both cases, the norm principle holds for all semisimple groups over $\overline{K}_\nu$. So if Question 2.4 has affirmative answer for K_ν , then the norm principle holds for all simply connected, and hence all semisimple, groups over K_ν (for all $\nu \in \Omega$). In particular, the norm principle holds for G over each K_ν . Since the kernel of the map $\rho_{G,\Omega}$ is trivial, the result follows from Lemma 2.1. $\square$

The norm principle is related to Serre's injectivity question. In [23, Page 233], Serre raised the following question: let K be a field, and let G be a smooth connected reductive K -group. Consider a set of finite field extensions L_i/K , and assume that the greatest common divisor of the degrees $[L_i : K]$ is 1. Consider the natural map

$$H^1(K, G) \longrightarrow \prod_i H^1(L_i, G)$$

induced by restriction. Does this map have trivial kernel?

In [3, Theorem 1.2], Bhaskhar proved that if $\text{char } K \neq 2$, and if the Dynkin diagram of G contains connected components only of type A_n , B_n or C_n , then Serre's question has a positive answer for G . Using our results in this paper on the norm principle for groups of type D_n , we extend Bhaskhar's result to the case where the components can also be of type D_n (D_4 non-trialitarian) over some function fields.

Theorem 2.7. *Let K, G , and L_i 's be as above. Suppose the Dynkin diagram of G has components of types A_n , B_n , C_n , and D_n (D_4 non-trialitarian). Suppose one the following holds:*

- *K is function field of a smooth projective geometrically integral curve over a p -adic field, or*
- *K is function field of a smooth projective geometrically integral curve over a number field k , and G is defined over k .*

Then, Serre's question has an affirmative answer for G .

Proof. By [3, Lemma 3.4], it suffices to prove the norm principle for simple components of the semisimple part of G . If a simple component has one of types A_n , B_n , or C_n , then we know the norm principle holds for that component by the main theorem in [1]. For components of type D_n (D_4 non-trialitarian), the result follows from Corollary 1.3 and Theorem 2.3. $\square$

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